Excel SQL
Data Processing Studio

Complete User Guide

Hindi + English Step-by-Step Instructions

- Import Excel, CSV, JSON, XML Files
- Connect SQL Server, MySQL, PostgreSQL, SQLite
- VLOOKUP, XLOOKUP, Multi-XLOOKUP, Merge VLOOKUP
- Table Compare, Duplicate Finder, Fuzzy Matching
- Visual Query Builder + SQL Editor
- Export to Excel, CSV, JSON, XML, Parquet

Powered by DuckDB • PySide6 • Pandas • RapidFuzz

Version 1.0.0

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1. System Requirements & Installation

English: Before running the application, install required Python packages.

Hindi: App chalane se pehle ye packages install karo.

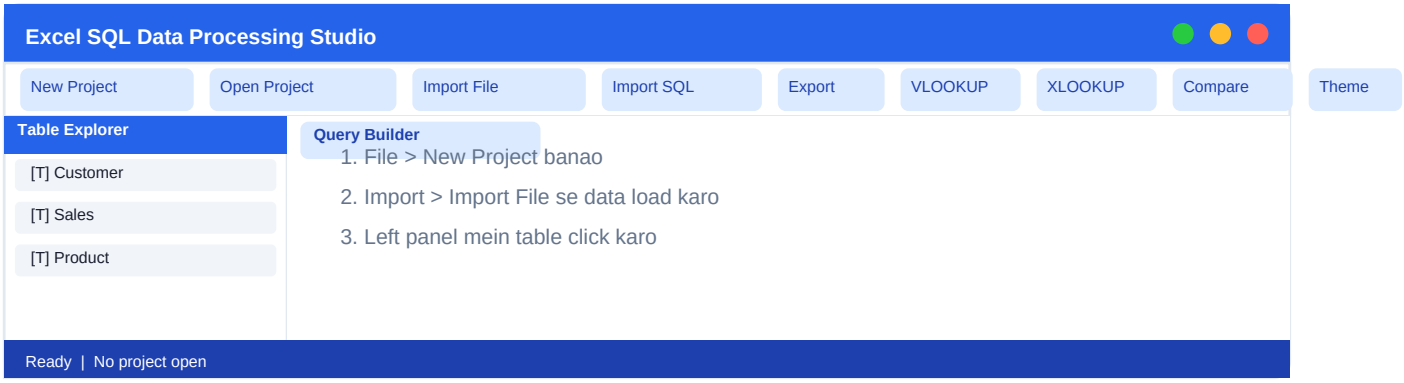
Package	Purpose / Kaam	Install Command
PySide6	Application UI Framework	pip install PySide6
duckdb	Local SQL Database Engine	pip install duckdb
pandas	Data Processing Library	pip install pandas
openpyxl	Excel File Support	pip install openpyxl
sqlalchemy	SQL Database Connector	pip install sqlalchemy
rapidfuzz	Fuzzy Matching Library	pip install rapidfuzz
pyodbc	SQL Server Driver (optional)	pip install pyodbc
pymysql	MySQL Driver (optional)	pip install pymysql
psycopg2-binary	PostgreSQL Driver (optional)	pip install psycopg2-binary

✓ **TIP** Run_App.bat double-click karo - ye sab automatically install karegi!

2 First Launch - Create New Project

Pehli Baar App Kholna

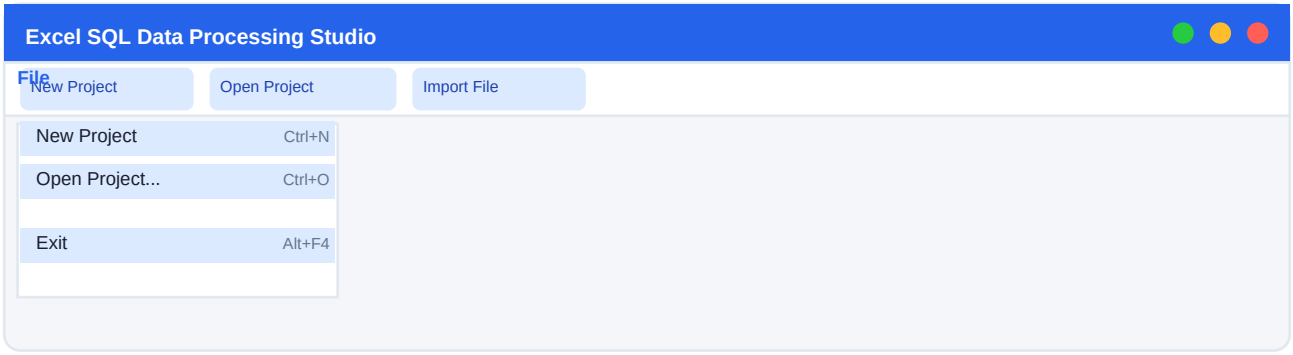
What you see when app opens / App kholne par kya dikhta hai



Layout Explanation / Layout Ki Jankari

Area / Jagah	English	Hindi
Top Bar (Blue)	Menu bar with File, Import, Export, Tools	Upar neela bar - File, Import, Export, Tools menu
Toolbar	Quick action buttons	Jaldi kaam karne ke buttons
Left Panel	Table Explorer - shows all imported tables	Left mein Table Explorer - sab tables dikhata hai
Center Area	Data Grid - shows table data	Center mein Data Grid - data dikhata hai
Bottom Bar	Status bar - shows current status	Neeche status bar - current status dikhata hai

Step: Create New Project / Naya Project Banao

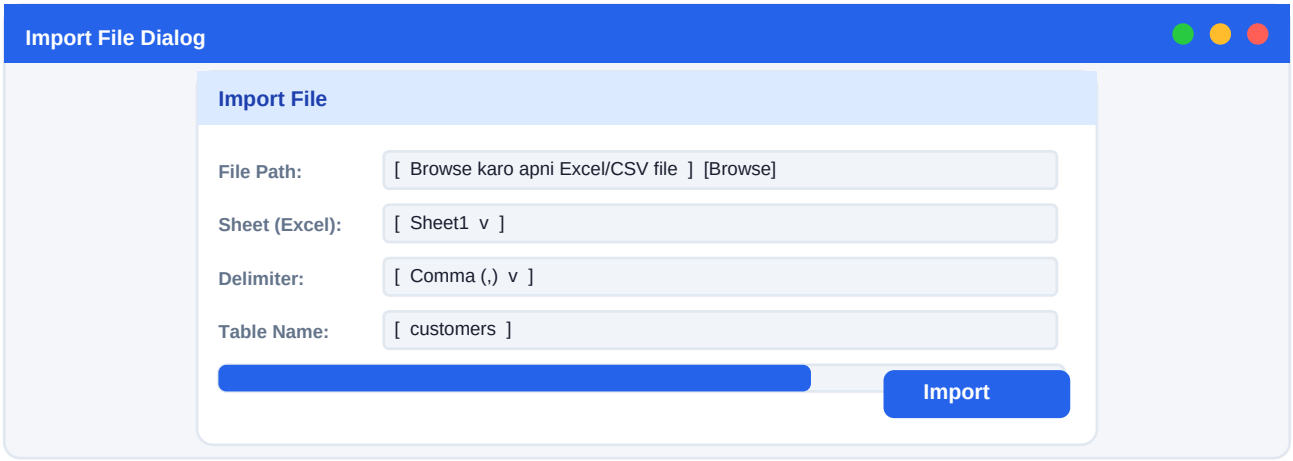


- **Step 1:** Click on File menu (top left) / File menu pe click karo (upar left mein)
- **Step 2:** Click 'New Project' / 'New Project' pe click karo
- **Step 3:** Choose a folder and type project name e.g. MyProject.duckdb / Folder choose karo aur naam likho jaise MyProject.duckdb
- **Step 4:** Click Save / Save button dabao
- **Step 5:** Left panel will show the database name / Left panel mein database naam dikhega

■ **INFO** Project file .duckdb format mein save hoti hai. Ise doosri jagah move kar sakte ho.

3 Import Excel / CSV / JSON / XML File

File Import Karna



How to Import a File / File Kaise Import Kare

- **Step 1:** Click menu: Import > Import File... / Menu mein Import > Import File... click karo
- **Step 2:** Click 'Browse' button / Browse button click karo
- **Step 3:** Select your Excel (.xlsx/.xls), CSV, JSON or XML file / Apni file select karo - Excel, CSV, JSON ya XML
- **Step 4:** For Excel: Select the sheet name from dropdown / Excel ke liye: Dropdown se sheet select karo
- **Step 5:** For CSV: Select delimiter (Comma/Tab/Pipe) / CSV ke liye: Delimiter select karo (Comma/Tab/Pipe)
- **Step 6:** Enter a table name (or use auto-filled name) / Table ka naam likho (ya auto-fill rehne do)
- **Step 7:** Click 'Import' button / Import button dabao
- **Step 8:** Progress bar dikhega - wait karo / Progress bar dikhega - rukke raho
- **Step 9:** Done! Table left panel mein dikhega / Ho gaya! Table left panel mein aa jayega

Supported File Formats / Supported Formats

Format	Extension	Notes / Dhyan de
Excel	.xlsx, .xls	Multiple sheets supported / Multiple sheets milti hain
CSV	.csv	Any delimiter - comma, tab, pipe / Koi bhi delimiter
TSV	.tsv	Tab-separated / Tab se alag
Text	.txt	Tab/comma separated / Tab ya comma se alag
JSON	.json	Array of objects format / Objects ka array
XML	.xml	Standard XML with rows / Standard XML

4 Import from SQL Database

SQL Database se Import Karna

Connection Details / Connection Jankari

- **Step 1:** Click 'Import' > Import from SQL Database / Import > Import from SQL Database click karo
- **Step 2:** Select Database Type (SQL Server/MySQL/PostgreSQL/SQLite) / Database type select karo
- **Step 3:** Enter Host (e.g. localhost or server IP) / Host bharo (jaise localhost ya server IP)
- **Step 4:** Enter Port (SQL Server=1433, MySQL=3306, PostgreSQL=5432) / Port bharo
- **Step 5:** Enter Database name / Database ka naam bharo
- **Step 6:** Enter Username and Password / Username aur Password bharo
- **Step 7:** Click 'Test & Connect' / Test & Connect button dabao
- **Step 8:** Table list dikhegi - select karo jo chahiye / Tables ki list aayegi - jo chahiye select karo
- **Step 9:** For Incremental: Check 'Incremental Import' and enter Key Column / Incremental ke liye: Checkbox lagao aur Key Column bharo
- **Step 10:** Click 'Import Selected' / Import Selected dabao

Database	Default Port	Required Driver / Zarooratan
SQL Server	1433	pip install pyodbc + ODBC Driver 17
MySQL	3306	pip install pymysql
PostgreSQL	5432	pip install psycopg2-binary
SQLite	N/A	Built-in - koi driver nahi chahiye

■ **INFO** Copy Data Mode: Data local DuckDB mein copy hota hai. Original database mein koi change nahi hota.

5 Viewing Data in Grid

Data Grid mein Data Dekhna

Grid Features / Grid Ki Suvidhaen

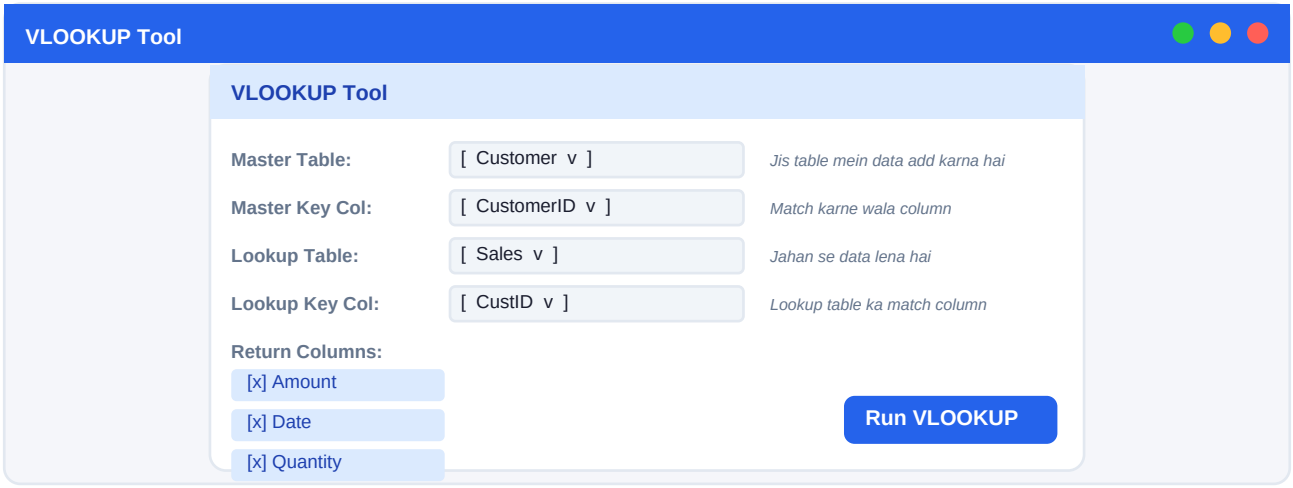
Feature / Button	English	Hindi
Click table name (left panel)	Left panel mein table name click karo	Table data center mein dikhega
Search box	Koi bhi text type karo search box mein	Sab columns mein search hoga
Column header click	Column header pe click karo	Column ke hisaab se sort hoga
Right-click on grid	Grid mein right-click karo	Copy, Hide Column, Freeze options milenge
Page size dropdown	Page size select karo (500/1000/5000)	Ek baar mein kitne rows dikhne hain
Prev / Next buttons	Prev aur Next buttons use karo	Pages ke beech navigate karo

6 VLOOKUP Tool

VLOOKUP Tool

English: VLOOKUP matches rows from one table to another and brings selected columns.

Hindi: VLOOKUP ek table ke rows ko doosri table se match karke columns laata hai.



- **Step 1:** Click: Tools > VLOOKUP / Menu mein Tools > VLOOKUP click karo
- **Step 2:** Select Master Table (main table jisme data add karna hai) / Master Table select karo
- **Step 3:** Select Master Key Column (matching column) / Master Key Column select karo - ye match hoga
- **Step 4:** Select Lookup Table (jahan se data lena hai) / Lookup Table select karo
- **Step 5:** Select Lookup Key Column (lookup table ka matching column) / Lookup Key Column select karo
- **Step 6:** Select Return Columns (jo columns laane hain - multiple select kar sakte ho) / Return Columns select karo - multiple select karo
- **Step 7:** Enter Output Table name / Output Table ka naam likho
- **Step 8:** Click 'Run VLOOKUP' / Run VLOOKUP button dabao
- **Step 9:** Result table left panel mein aa jayega / Result table left panel mein dikhega

✓ **TIP** Example: Customer table + Sales table ko CustomerID se join karo aur Amount, Date columns lo.

7 XLOOKUP Tool

XLOOKUP Tool

English: XLOOKUP is like VLOOKUP but supports both exact and contains (partial) matching.

Hindi: XLOOKUP VLOOKUP jaisa hai lekin partial match bhi kar sakta hai.

Feature	VLOOKUP	XLOOKUP
Match Type	Exact only	Exact + Contains (partial)
Use Case	ID se match karo	Name se partial match bhi kar sakte ho
Example	CustomerID = 'C001'	Name CONTAINS 'Sharma'

Steps same as VLOOKUP + choose Match Type:

- Select **Exact Match** - exact ID/code match ke liye
- Select **Contains Match** - jab naam mein kuch word ho tab match karo

✓ **TIP** Contains match useful hai jab 'Tata Motors' aur 'TATA MOTORS LTD' ko same maanna ho.

8 Multi-XLOOKUP Tool

Multi-XLOOKUP Tool

English: Join multiple lookup tables to master table in one single operation.

Hindi: Ek baar mein kai lookup tables ko master table se join karo.

- **Step 1:** Click: Tools > Multi-XLOOKUP / Tools > Multi-XLOOKUP click karo
- **Step 2:** Select Master Table / Master Table select karo
- **Step 3:** Select Master Key column / Master Key column select karo
- **Step 4:** Click '+ Add Lookup Table' for each lookup / Har lookup ke liye '+ Add Lookup Table' click karo
- **Step 5:** For each entry: select Table, Key column, Return columns, Prefix / Har entry mein: Table, Key, Return columns, Prefix set karo
- **Step 6:** Enter Output table name / Output table naam likho
- **Step 7:** Click 'Run Multi-XLOOKUP' / Run Multi-XLOOKUP dabao

✓ TIP Example: Customer + Sales + Address + Product - sab ek saath join karo!

9 Merge VLOOKUP Tool

Merge VLOOKUP Tool

English: When one ID has multiple matching rows, Merge VLOOKUP pivots them into separate columns.

Hindi: Jab ek ID ke liye lookup table mein kai rows hon, unhe alag columns mein daalta hai.

Example / Udaharan

Before (Lookup Table):

CustomerID	Mobile (Lookup Table)
C001	9876543210
C001	8765432109
C001	7654321098
C002	6543210987

After Merge VLOOKUP (Result):

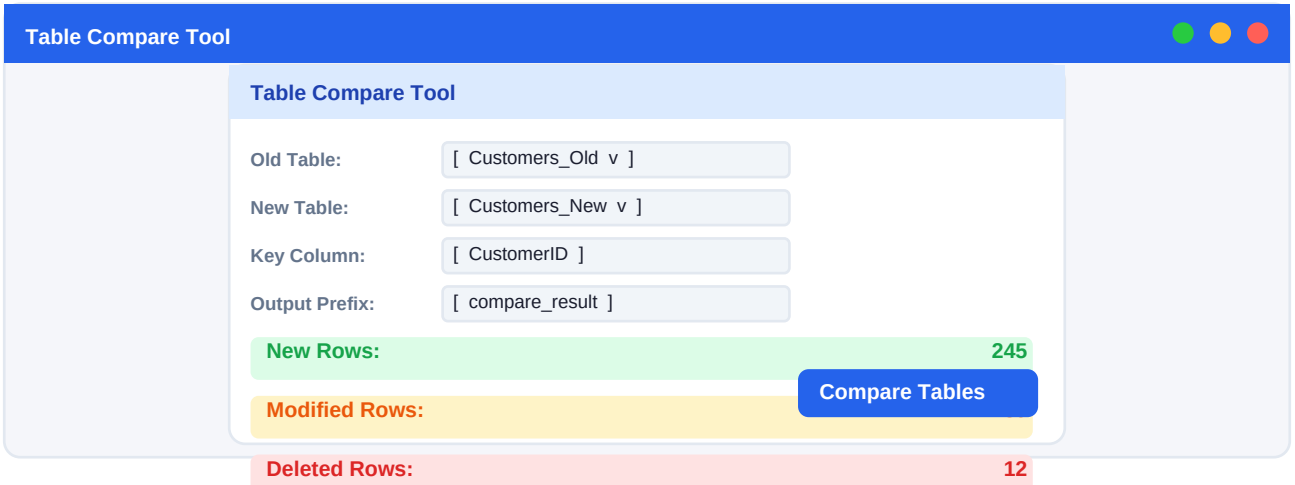
CustomerID	Mobile_M1	Mobile_M2	Mobile_M3
C001	9876543210	8765432109	7654321098
C002	6543210987		

10 Table Compare Tool

Table Compare Tool

English: Compare old and new versions of a table to find new, modified, and deleted records.

Hindi: Do tables compare karo - naye, badle hue, aur delete hue records dhundho.



- **Step 1:** Import both Old and New table versions / Dono tables import karo - purani aur nayi
- **Step 2:** Click: Tools > Table Compare / Tools > Table Compare click karo
- **Step 3:** Select Old Table / Old Table select karo
- **Step 4:** Select New Table / New Table select karo
- **Step 5:** Enter Key Column (unique ID column like CustomerID) / Key Column bharo - jaise CustomerID
- **Step 6:** Enter Output Prefix (e.g. 'compare') / Output Prefix likho jaise 'compare'
- **Step 7:** Click 'Compare Tables' / Compare Tables button dabao

Output Tables Created / Jo Tables Banegi

Table Name	Contains / Kya hoga
compare_new	Records in new table but not in old / Naye records
compare_modified	Records changed between old and new / Badle hue records
compare_deleted	Records in old but removed from new / Delete hue records

11 Duplicate Finder

Duplicate Finder

English: Find duplicate records based on one or more columns.

Hindi: Ek ya kai columns ke basis par duplicate records dhundho.

- **Step 1:** Click: Tools > Duplicate Finder / Tools > Duplicate Finder click karo
- **Step 2:** Select the table to check / Table select karo jisme check karna hai
- **Step 3:** Select one or more columns to check duplicates / Columns select karo - ek ya kai
- **Step 4:** Enter Output table name / Output table naam likho
- **Step 5:** Click 'Find Duplicates' / Find Duplicates dabao
- **Step 6:** Result table mein sab duplicates honge with count / Result mein duplicates aur unki count dikhegi

✓ TIP Tip: Email column mein duplicate check karo - double entries pakad sakte ho.

12 Fuzzy Matching Tool

Fuzzy Matching Tool

English: Match records that are similar but not exactly equal. Uses RapidFuzz library.

Hindi: Aisi records match karo jo bilkul same nahi hain lekin similar hain.

- **Step 1:** Click: Tools > Fuzzy Matching / Tools > Fuzzy Matching click karo
- **Step 2:** Select Master Table and Master Column / Master Table aur Column select karo
- **Step 3:** Select Lookup Table and Lookup Column / Lookup Table aur Column select karo
- **Step 4:** Set Min Confidence % (default 80%) / Min Confidence % set karo (default 80%)
- **Step 5:** Set Max Matches per value (default 1) / Max Matches per value set karo
- **Step 6:** Enter Output table name / Output table naam likho
- **Step 7:** Click 'Run Fuzzy Match' / Run Fuzzy Match dabao

Match Value	Matched Value	Confidence Score
Reliance Ind.	Reliance Industries Ltd	96.5%
Tata Motors	Tata Motors Limited	94.2%
Infosys Tech	Infosys Technologies	91.8%

13 Query Builder & SQL Editor

Query Builder aur SQL Editor

English: Build SQL queries visually or write raw SQL. Results saved as new tables.

Hindi: SQL query visual builder se banao ya seedha SQL likho. Result nayi table mein save hoga.

Visual Builder Tab / Visual Builder Tab

Field / Box	Purpose / Kaam
FROM Table dropdown	Jis table se data chahiye select karo
SELECT Columns list	Kaunse columns dikhane hain - multiple select karo
JOIN Table	Doosri table join karni ho to select karo
ON field	Join condition likho: a.ID = b.ID
WHERE field	Filter condition likho: "Amount" > 1000
GROUP BY field	Group karne ke liye column likho
ORDER BY field	Sort karne ke liye likho: "Date" DESC
LIMIT field	Kitne rows chahiye - default 10000
Save As Table	Result table ka naam likho
Preview SQL button	SQL dekhne ke liye - koi action nahi
Run Query button	Query chalao - result table banegi

SQL Editor Tab / SQL Editor

- SQL Editor tab mein click karo
- Seedha SQL query likho
- Save result as: mein output table naam likho
- Execute SQL button dabao

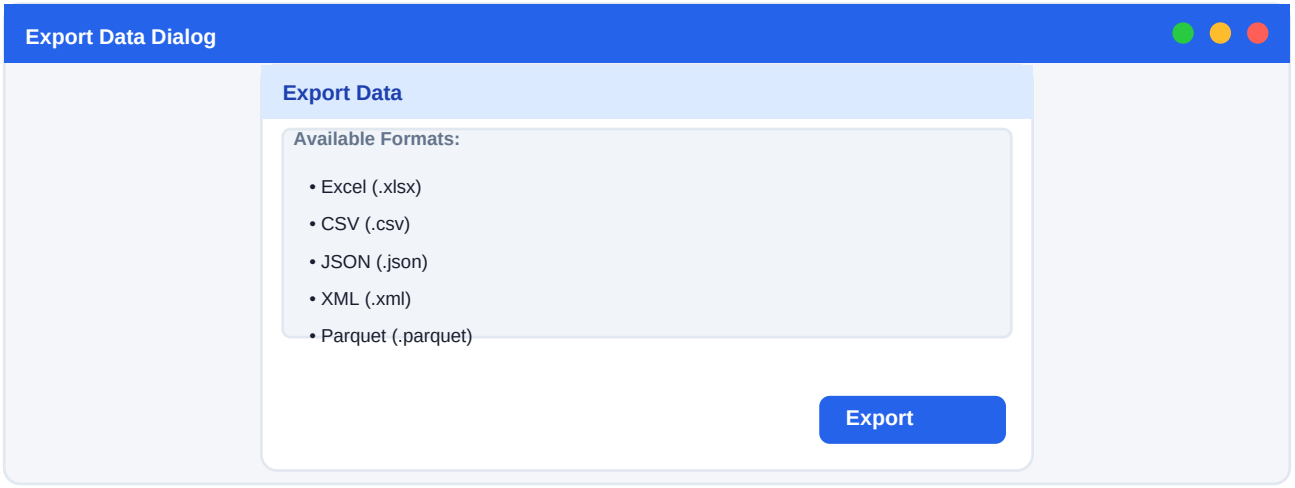
■ **INFO** SQL Editor mein koi bhi DuckDB SQL likh sakte ho - SELECT, CREATE, INSERT, etc.

14 Export Data

Data Export Karna

English: Export any table to Excel, CSV, JSON, XML, or Parquet format.

Hindi: Kisi bhi table ko Excel, CSV, JSON, XML ya Parquet mein export karo.



- **Step 1:** Click: Export > Export Table... (or toolbar Export button) / Export > Export Table... click karo
- **Step 2:** Select the table to export / Export karne wali table select karo
- **Step 3:** Select format: Excel, CSV, JSON, XML, or Parquet / Format select karo: Excel, CSV, JSON, XML ya Parquet
- **Step 4:** Click 'Browse' to choose save location / Browse click karo - save location choose karo
- **Step 5:** Click 'Export' button / Export button dabao
- **Step 6:** File save ho jayegi! / File save ho jayegi!

Format Comparison / Format Comparison

Format	Best For / Kab Use Karo	File Size
Excel (.xlsx)	Office users, formatting chahiye	Medium
CSV (.csv)	Any software, universal format	Small
JSON (.json)	Web apps, APIs ke liye	Medium
XML (.xml)	Legacy systems, structured data	Large
Parquet (.parquet)	Big data, fastest for large files	Smallest (compressed)

15 Toolbar Quick Reference

Toolbar Quick Reference

All toolbar buttons and their functions / Sab toolbar buttons aur unka kaam:

Button	English	Hindi
New Project	Create new .duckdb project file	Naya project file banao
Open Project	Open existing project	Purana project kholo
Import File	Import Excel/CSV/JSON/XML	File import karo
Import SQL	Connect & import from SQL database	SQL database se import karo
Export	Export current table	Current table export karo
VLOOKUP	Open VLOOKUP tool	VLOOKUP tool kholo
XLOOKUP	Open XLOOKUP tool	XLOOKUP tool kholo
Multi-XLookup	Open Multi-XLOOKUP tool	Multi-XLOOKUP tool kholo
Merge VLookup	Open Merge VLOOKUP tool	Merge VLOOKUP tool kholo
Compare	Open Table Compare tool	Table Compare tool kholo
Duplicates	Open Duplicate Finder	Duplicate Finder kholo
Fuzzy Match	Open Fuzzy Matching tool	Fuzzy Matching tool kholo
Theme	Toggle Dark/Light mode	Dark/Light mode badlo

Right-Click Menu (on data grid) / Right-Click Menu

Option	English	Hindi
Copy Selected	Copy selected rows to clipboard	Selected rows clipboard mein copy karo
Copy All	Copy entire page to clipboard	Puri page clipboard mein copy karo
Hide Column	Hide the clicked column	Column chhupao
Freeze to Here	Freeze columns up to this point	Column yahan tak freeze karo
Refresh	Reload table data	Data refresh karo

■ INFO Log files yahan milenge: C:\Users\YourName\.excel_sql_studio\logs\app.log

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